

Filippo Lollato

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EXPERIENCE

ML Student

University of Trento

2024 — present

Trento, Italy

- Master's level coursework and projects spanning deep learning, NLP, computer vision, reinforcement learning, and autonomous systems.
- Lead architect on the ASA Autobots multi-agent capstone.

University Tutor

University of Trento

2024 — present

Trento, Italy

- Tutoring undergraduate students in algorithms, data structures, and foundational programming.
- Mentoring students to make complex technical ideas accessible and building leadership skills.

EDUCATION

Master's Degree in Artificial Intelligence Systems

University of Trento

2024 — Present

Trento, Italy

Bachelor's Degree in Computer Science

University of Trento

2021 — 2024

Trento, Italy

- Final Grade: [GRADE/110L]

PROJECTS

kindle — Machine Learning Framework

A backend-agnostic framework for building neural networks in Rust.

Rust

- A Machine Learning framework built in Rust which aims to provide the ability to check all tensor invariants at compile time while still having the extreme flexibility and ergonomics of modern frameworks.
- Achieved compile-time ML shape & type safety with zero runtime overhead.

evolcpp — Neuroevolution Library

A C++ port of kindle implementing its core abstractions.

C++, CMake

- Implemented complex template meta-programming to achieve robust compile-time shape safety principles matching the Rust counterpart.

ASA Autobots — Multi-Agent System

Cooperative delivery system combining LLMs and symbolic planning.

Node.js, PDDL, BDI, LLM

- Engineered a system where BDI executors, an LLM coordinator (Llama 3.3 70B), and a PDDL planner collaborate in real time.
- Acted as Co-lead for a team of 2, achieving ~91.5% test coverage with a robust 3-layer architecture.

signal_image_video — Multi-Object Tracking

3D camera-LiDAR fusion tracking system.

Python, PyTorch, Kalman Filter

- Built a tracking system combining a CTRV-EKF Kalman filter with a Hungarian algorithm assignment step.
- Trained and evaluated on the KITTI benchmark, achieving reduced RMSE over the linear tracking baseline.

NLU — Intent & Slot Filling

Python, PyTorch, BERT

Natural Language Understanding on the ATIS dataset.

- Built the training pipeline from scratch, fine-tuning BERT with a joint intent classification and slot-filling head.
- Achieved 0.9765 intent accuracy and high slot F1 score.

TECHNICAL SKILLS

Languages: Rust, C++, Python, TypeScript, JavaScript, SQL

Machine Learning Stack: PyTorch, Jupyter, Ollama, HuggingFace

Deployment & Operations: Docker, ONNX Runtime, Axum, REST APIs, Git, Linux

Web Technologies: React, Astro, Node.js, TailwindCSS